

CBDCs: Could the Biggest Impact Occur Behind the Scenes?

The debate around CBDCs often focuses on consumers, but some of the most significant changes may occur within the financial infrastructure used by banks to transfer and settle money.



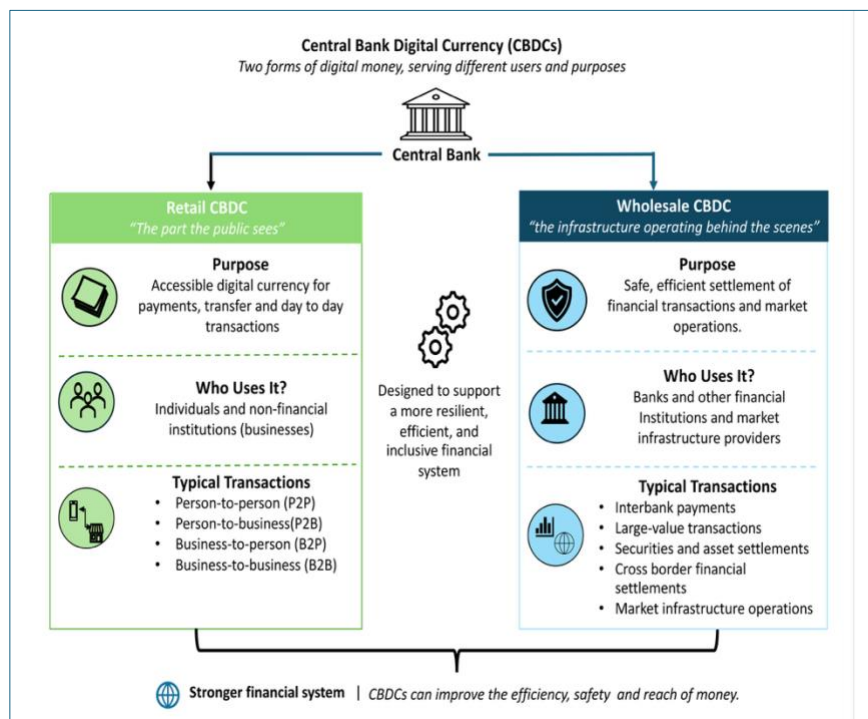
1. Introduction

The public conversation around Central Bank Digital Currencies (CBDCs) has increasingly focused on their potential implications for consumers, particularly with respect to privacy, government surveillance, programmability, and the future role of cash. Central banks, meanwhile, have emphasised potential benefits including faster and more efficient payments, greater financial inclusion, and improved cross-border transactions.

Yet, beyond these considerations another important development is taking place behind the scenes. Central banks alongside other financial institutions are exploring how digital forms of central bank money could reshape the financial system, particularly, how money is transferred and financial transactions are settled.

According to the Bank for International Settlements (BIS), retail CBDCs (rCBDCs) - designed for use by individuals and businesses - remain the more widely pursued application, particularly among emerging market and developing economies. Nonetheless, wholesale CBDCs, designed for use by financial institutions, are gaining

Figure 1: Retail and wholesale CBDCs: different users, different roles



attention as a potential means of improving the transfer and settlement of large-value transactions. The findings of the BIS's 2024 survey, as reported by Illes, Kosse and Wiertz (2025)¹, showed that 91 percent of the central banks surveyed were exploring retail CBDCs, wholesale CBDCs, or both, with wholesale CBDC work progressing to more advanced stages than work on retail CBDC.

This growing focus on wholesale CBDCs raises an important question: “Could the most significant impact of CBDCs occur not in the money people use every day, but in the financial infrastructure operating behind it?”

2. Why the Growing Interest in Wholesale CBDCs?

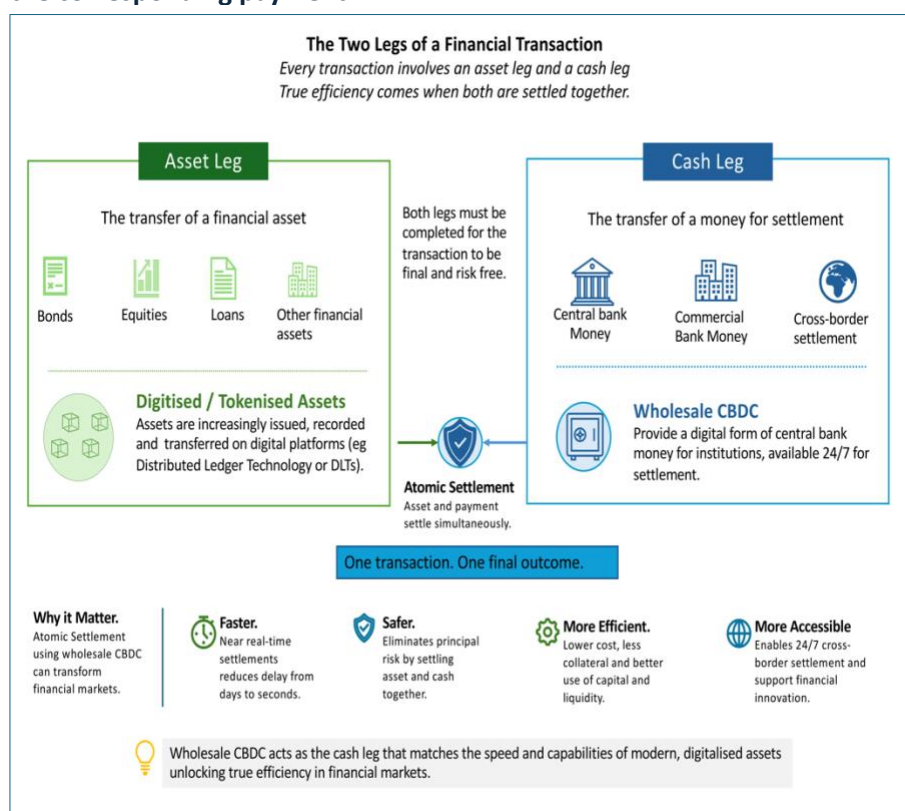
Why are central banks and financial institutions exploring wholesale CBDCs? One reason is the growing need for safer and more efficient ways to settle large-value financial transactions, particularly as financial markets become increasingly digital.

A financial transaction typically involves two components: the asset leg, which is the financial asset being transferred, such as a bond or security, and the cash leg, which is the money used to pay for it. As financial institutions are increasingly exploring ways to digitise and automate the asset leg, the systems used to transfer and settle the corresponding cash must also evolve.

Currently, traditional settlement arrangements can involve multiple intermediaries and processes that take time, particularly for cross-border transactions. These arrangements can increase costs, tie up liquidity, and expose participants to settlement risk - the possibility that one side of a transaction is completed while the other side is delayed or fails.

Private digital forms of money, including stablecoins, offer one possible way to transfer value digitally. However,

Figure 2: Financial transactions involve both the transfer of an asset and the corresponding payment.



¹ Anamaria Illes, Anneke Kosse and Peter Wiertz, Advancing in tandem: Results of the 2024 BIS survey on central bank digital currencies and crypto, BIS Papers No. 159, Bank for International Settlements, August 2025.

stablecoins are private-sector liabilities and can introduce risks related to the issuer, the assets backing the stablecoin, and its ability to maintain a stable value. Wholesale CBDCs offer a different approach by using central bank money for settlement between financial institutions.

One example is **Project mBridge**, a collaboration involving the Bank for International Settlements (BIS) and the central banks and monetary authorities of China, Hong Kong SAR, Thailand, and the United Arab Emirates. The project explored how a shared platform could enable participating financial institutions to conduct cross-border, multi-currency transactions using central bank money.

The broader objective of wholesale CBDCs is therefore not simply to create another form of digital money. Rather, it is to explore whether central bank money can be integrated into modern financial infrastructure to make the settlement of financial transactions faster, safer, and more efficient, while reducing some of the costs, delays, and risks associated with existing arrangements.

3. How Wholesale CBDCs Could Change Payment Infrastructure

The interest in wholesale CBDCs is closely linked to the potential to improve the infrastructure used to transfer and settle money and financial assets. By introducing digital central bank money into financial market infrastructure, wholesale CBDCs could make settlement faster, more efficient, and better suited to increasingly digital financial markets.

Some of the potential benefits include:

- **Atomic settlement:** The transfer of an asset and the corresponding payment can be coordinated so that one does not occur without the other, reducing settlement risk.
- **Faster settlement:** Transactions could be settled more quickly, reducing the time that funds and assets remain in transit and potentially freeing up liquidity.
- **Greater availability:** Digital settlement infrastructure could support transactions beyond traditional banking hours, potentially enabling more continuous cross-border activity.
- **Improved data:** Modern payment standards, such as ISO 20022, allow payments to carry more structured information, which can support automation, compliance, and transaction monitoring.

Together, these developments could make the underlying financial infrastructure more efficient and better suited to increasingly digital financial markets. The objective is therefore not simply to create another form of digital money, but to improve the infrastructure through which money and financial assets are transferred and settled.

4. Who Else Can Benefit from Wholesale CBDCs?

The potential benefits of wholesale CBDCs are not limited to central banks and financial institutions. If wholesale CBDC infrastructure makes the transfer and settlement of money faster, safer, and more efficient, some of these benefits could extend to businesses and consumers through the financial institutions and payment networks they use.

Businesses: Small and medium-sized enterprises (SMEs), particularly those engaged in international trade, could benefit from faster and more efficient cross-border payments. International transactions can involve multiple intermediaries, fees, processing delays, and liquidity requirements, making it more

difficult for businesses to manage working capital and pay overseas suppliers efficiently. If wholesale CBDC infrastructure reduces some of these frictions, commercial banks could potentially pass on some of the benefits through faster settlement, lower transaction costs, and improved payment services.

For an SME importing goods from overseas, for example, faster settlement could provide quicker access to funds and greater certainty about when a payment has been completed. The SME would not use a wholesale CBDC directly; rather, it could benefit from improvements in the financial infrastructure supporting the services provided by its bank.

Consumers: Consumers could also benefit indirectly if improvements in wholesale payment infrastructure allow financial institutions to provide faster, more reliable, and less costly payment services. Although consumers would generally interact with retail payment systems rather than wholesale CBDCs themselves, the infrastructure supporting those services could become more efficient.

The extent to which these benefits reach businesses and consumers will depend on how wholesale CBDC systems are designed, how widely they are adopted, and how effectively they connect with existing payment and financial market infrastructure.

5. Conclusion

The most significant impact of CBDCs may not necessarily come from a new payment app or from consumers replacing cash with a digital currency. Instead, it may emerge from changes to the infrastructure that supports financial transactions.

Wholesale CBDCs could provide a new way for financial institutions to settle transactions using central bank money, potentially making settlements faster, reducing settlement risks, and supporting greater integration between digital financial assets and payment systems.

For consumers and businesses, these changes may not be immediately visible. Everyday banking applications could look much the same, even as the systems supporting them becomes more digital, interconnected, and efficient.

This is why the CBDC debate should extend beyond the question of whether people will use a digital currency in their daily lives. An equally important question is whether **digital forms of central bank money can improve the underlying systems through which the financial system transfers and settles money and assets.**

If that happens, the most significant impact of CBDCs may indeed occur behind the scenes.